Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [14EE71]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2018

POWER SYSTEM ANALYSIS-II

Time: 3 Hours **Maximum Marks:70**

Instructions: i. Answer any **FIVE** full Questions choosing at least two from each part.
ii. Missing data may be suitably assumed.

PART - A

1. a. Obtain the primitive impedance and admittance form of an element. Draw their circuit representations. **07**

Consider three passive elements whose data is as follows. Form the primitive impedance and admittance matrices.

Element	z_k (pu)	Mutual impedance
1	j 0.6	-
2	j 0.5	j 0.2 (with element 1)
3	j 0.7	-

b. The reduced branch-node incidence matrix of a network is given by **07**

$$Ar(b\times n)=\begin{bmatrix}1 & 0 & 0\\ -1 & 1 & 0\\ 0 & 1 & 0\\ 0 & -1 & 1\\ 0 & 0 & 1\end{bmatrix}$$

Draw the oriented graph. Draw a tree, corresponding co-tree, Basic cut-set incidence matrix and Basic loop incidence matrix. What are the describing equations for the two matrices?

2. a. The line data of a system is follows: **07**

Line	R	X
1-2	0.05	0.15
1-3	0.1	0.3
2-3	0.15	0.45
2-4	0.10	0.30
3-4	0.05	0.15

Formulate Y_{bus} without line 1-2. If line 1-2 is added, show how Y_{bus} is modified.

b. Obtain the model of a transformer connected between buses p-q with an off-nominal turns ration a:1. **07**

A transformer is connected between buses 4-3, with an impedance 0.0 + j0.133 pu, and turns ratio 0.909. Determine the elements of Ybus that would be modified with this.

3. a. Discuss the various operating constraints in solving load flow equations. **04**

b. The bus data of a system is given by **10**

Bus No.	P_i (pu)	Q_i (pu)	V_i
1	–	–	$1.04\angle 0^0$
2	0.5	– 0.2	–
3	– 1.0	0.5	–
4	– 0.3	– 0.1	–

The bus admittance matrix of the system is given by

$$Y_{bus} = \begin{bmatrix} 3.0 - j9.0 & -2.0 + j6.0 & -1.0 + j3.0 & 0.0 + j0.0 \\ -2.0 + j6.0 & 3.666 - j11.0 & -0.666 + j2.0 & -1.0 + j3.0 \\ -1.0 + j3.0 & -0.666 + j2.0 & 3.666 - j11.0 & -2.0 + j6.0 \\ 0.0 + j0.0 & -1.0 + j3.0 & -2.0 + j6.0 & 3.0 - j9.0 \end{bmatrix}$$

Obtain the load flow solution at the end of the first iteration for following cases:

i. All buses except bus 1 are PQ Buses.

ii. Bus 2 is a PV bus whose voltage magnitude is specified as 1.04.

4. a. Derive the equations to be solved for NR method of load flow solution in polar coordinates. **08**

b. State the assumptions made in the FDLF load flow solution. Hence, obtain the equations to be solved. **06**

PART - B

5. a. Write the swing equation and express it as first order differential equations. Give the equations to solve it using RK method. **07**

b. A 50 Hz, synchronous generator having inertia constant H = 5.2 MJ/MVA and x_d' = 0.3 pu is connected to an infinite bus through a double circuit line. The reactance of the connecting HT transformer is 0.2 pu and reactance of each line is 0.4 pu. $|E_g|$ = 1.2 pu and $|V|$ = 1.0 pu and P_e = 0.8 pu. A three phase fault occurs at the middle of one of the transmission lines and is cleared in 6.25 cycles by isolating the faulted line. Obtain the values of δ(t) for a period of 0.25 secs using a step size of 0.05 sec , by Modified Euler's method. **07**

6. a. On what factors does the load power depend? Give the sensitivity to these parameters of some common loads. What are the different ways in which the static loads can be modelled? **08**

b. Discuss the modelling of thermostatic loads and induction motor loads to account for their dynamic behavior. **06**

7. a. Derive equation for the addition of a branch between an existing bus p and a new bus q, of a power system network. If bus p is a reference bus what how are the equations modified. **06**

b. The Zbus of a partial network is given by **08**

$$Z_{bus} = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} j0.1 & 0 & j0.1 \\ 0 & j0.15 & 0 \\ j0.1 & 0 & j0.5 \end{bmatrix} \end{array}$$

An element is added between the buses 1-2. Obtain the modified Zbus.

8. a. The fuel cost in \$/MWh for two plants is given by **07**

$$F1 = 450 + 5.5\ PG1 + 0.005\ PG1^2$$

$$F2 = 500 + b2\ PG2 + c2\ PG2^2$$

Where PG1 and PG2 are in MW.

i. The incremental cost of power, λ, is 8 \$/MWh when total demand is 500 MW. Determine optimum scheduling.

ii. The incremental cost of power is 10\$/MWh when total demand is 1200 MW. Determine optimum scheduling neglecting.

iii. Find the coefficients b2 and c2.

b. In a system with two plants, the incremental fuel costs are given by **07**

$$\frac{dF_1}{dP_{G1}} = 0.01P_{G1} + 20 \text{ Rs / MWh}$$

$$\frac{dF_2}{dP_{G2}} = 0.015P_{G2} + 22.5 \text{ Rs / MWh}$$

The system is running under optimal schedule with $P_{G1} = P_{G2} = 100$ MW.

If $\frac{\partial P_L}{\partial P_{G2}} = 0.2$, find the plant penalty factors and $\frac{\partial P_L}{\partial P_{G1}}$.